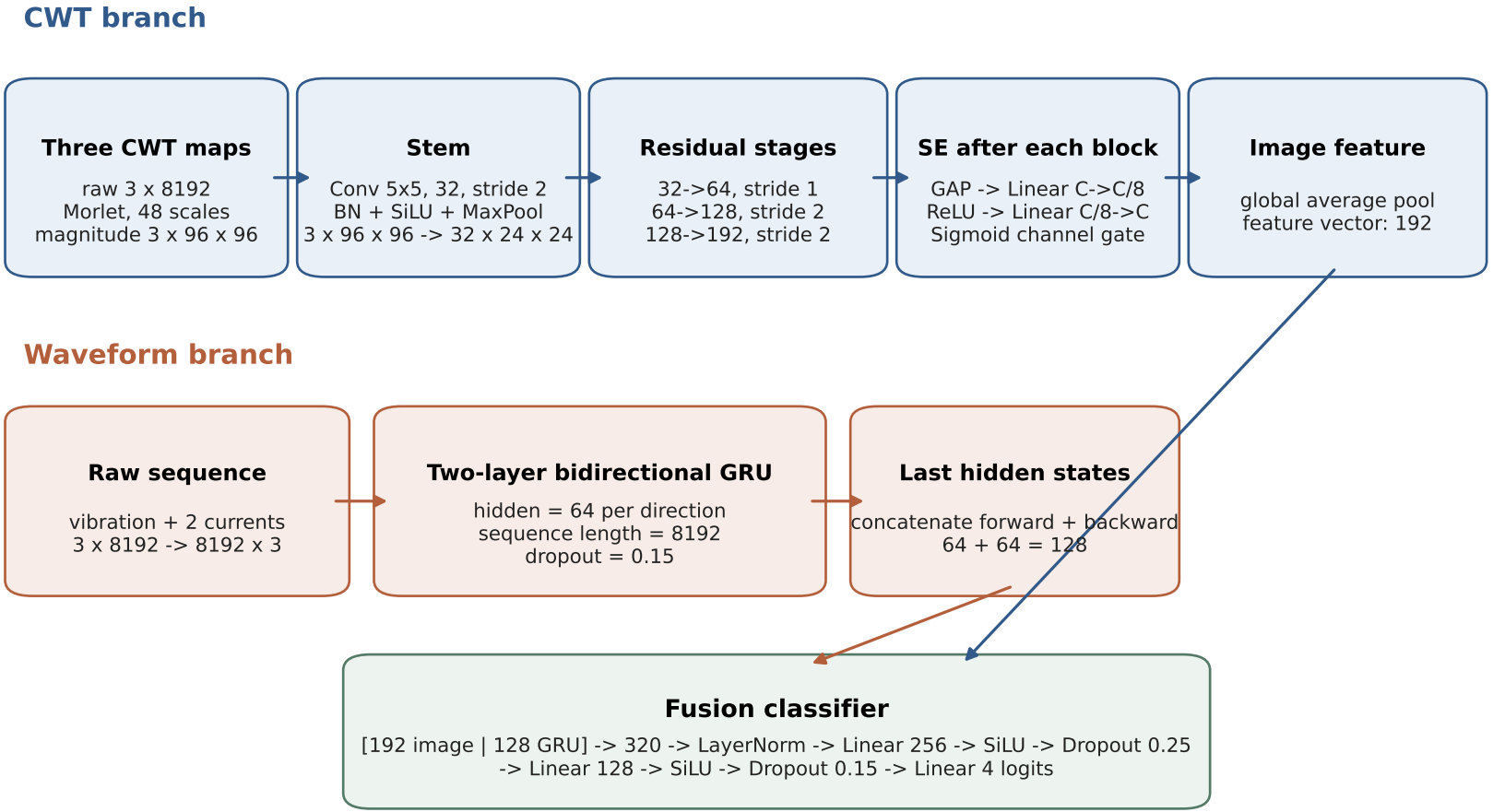
CWT branch
Three CWT maps
raw 3 x 8192
Morlet, 48 scales
magnitude 3 x 96 x 96
Stem
Conv 5x5, 32, stride 2
BN + SiLU + MaxPool
3 x 96 x 96 -> 32 x 24 x 24
Residual stages
32->64, stride 1
64->128, stride 2
128->192, stride 2
SE after each block
GAP -> Linear C->C/8
ReLU -> Linear C/8->C
Sigmoid channel gate
Image feature
global average pool
feature vector: 192
Waveform branch
Raw sequence
vibration + 2 currents
3 x 8192 -> 8192 x 3
Two-layer bidirectional GRU
hidden = 64 per direction
sequence length = 8192
dropout = 0.15
Last hidden states
concatenate forward + backward
64 + 64 = 128
Fusion classifier
[192 image | 128 GRU] -> 320 -> LayerNorm -> Linear 256 -> SiLU -> Dropout 0.25
-> Linear 128 -> SiLU -> Dropout 0.15 -> Linear 4 logits